

AI and Machine Learning Ethics

CM2010 Sociocultural Perspectives to Innovative Technologies

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Background

Sometimes this concept of ‘being ethical’ gets a bit of a tough time in the tech. community (whether that’s in industry or academia) because it can feel like a bit of a chore that might somehow restrain our work.

But a key purpose of today is for me to try and show you how, on the contrary, this leads to better tech and is pretty interesting to work on and consider.



What is ethics?

Ethics: Guiding behaviour based on moral principles

AI/ML ethics: concerns how developers, manufacturers, authorities and operators should behave in order to minimise the ethical risks that can arise from AI/ML in society, either from design, inappropriate application, or intentional misuse of the technology.

During this lecture we will focus on practical rather than philosophical ethics.

1. Ethical implications of AI/ML: bias, accountability, sustainability, surveillance, trust

2. How to develop AI/ML ethically : Mutual Shaping, Value Sensitive Design, Ethical Standards

Embracing ethics as an engineer

From the very beginning of a new project, let's:

1. Design, deploy and evaluate new tech using inherently responsible and ethical approaches

2. Consider the ethical implications and any associated risk from our tech.

3. Critically evaluate the real-world impact(s) (good and bad) of our tech.

How?

Practical things you can do, from today, to deliver more ethical AI:

1. Taking a **mutual shaping** view of, and approach to, technology and society.
2. Using **ethical standards** for best practice and ethical risk assessment.
3. Incorporating ethics into the **design stages** of technology

More later ☺

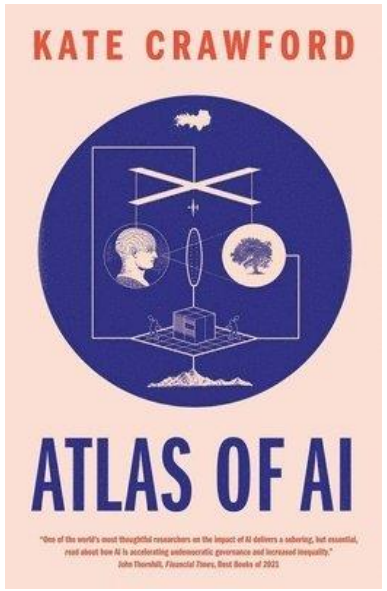
Standard vs Principle vs Framework

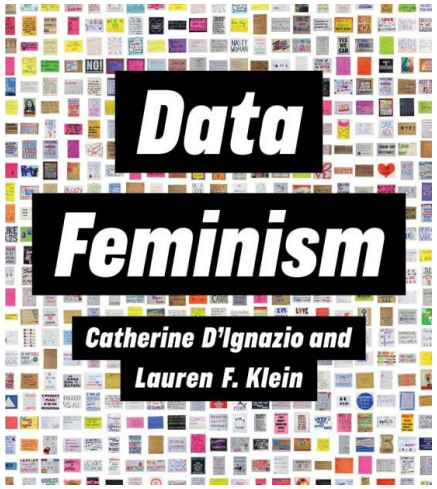
1. **Ethical Standards** are concrete rules or guidelines designed to govern behaviour within a specific context (e.g. GDPR).

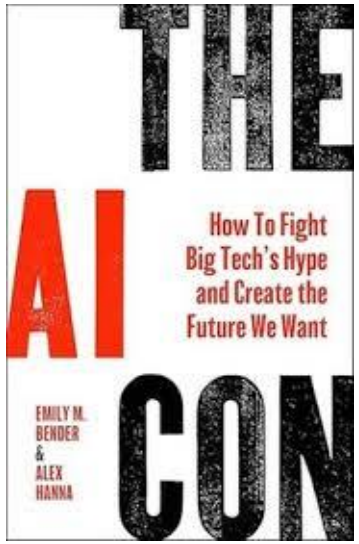
2. **Ethical Principles** are foundational moral values that underpin and motivate ethical standards and frameworks, guiding overall ethical behaviour (e.g. trust, fairness).

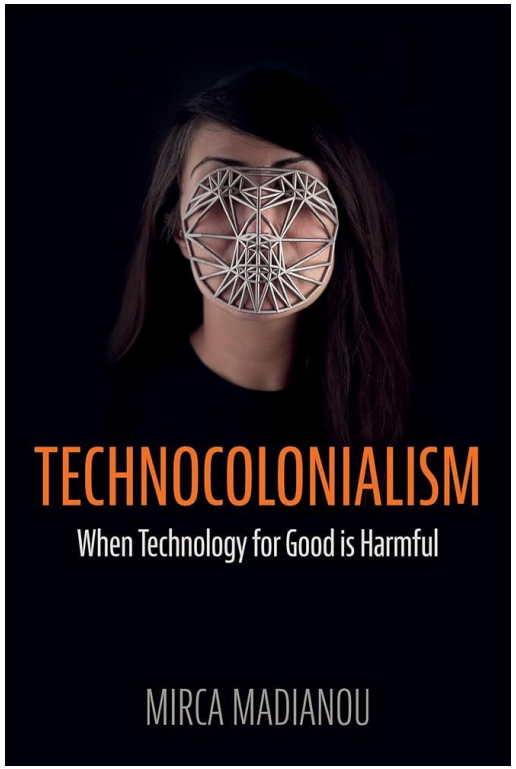
3. **Ethical Frameworks** provide structured approaches for thinking about and resolving ethical dilemmas, offering a methodology for ethical decision-making (e.g. Utilitarianism).

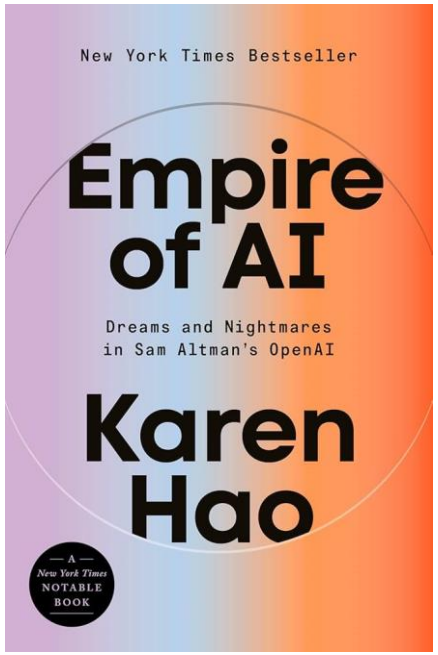
On my bookshelf

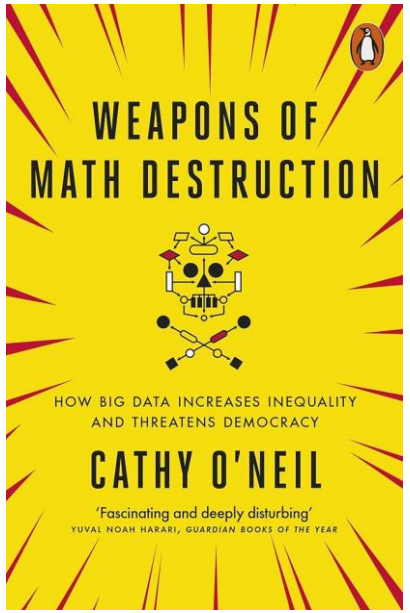


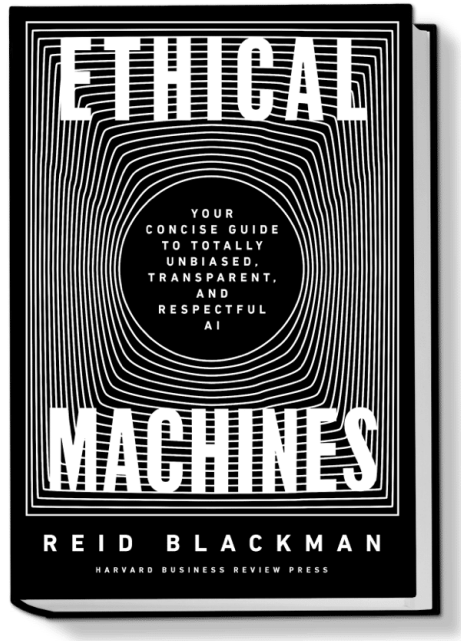








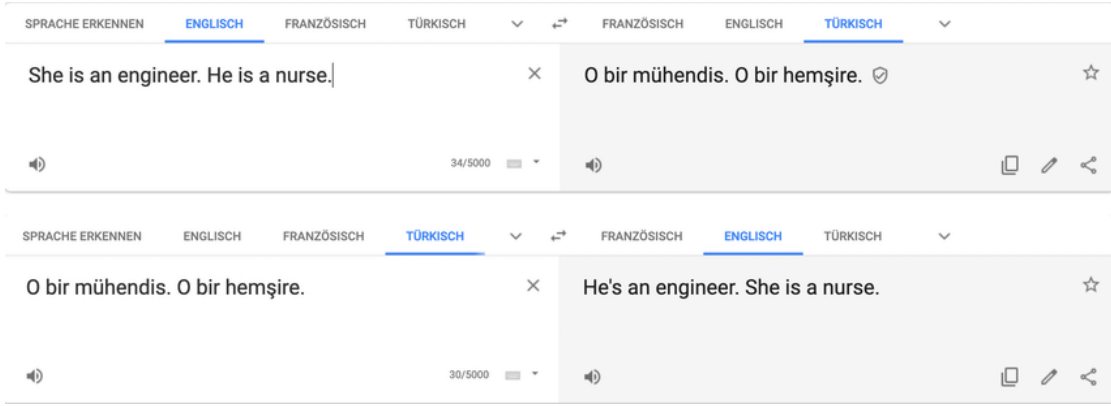


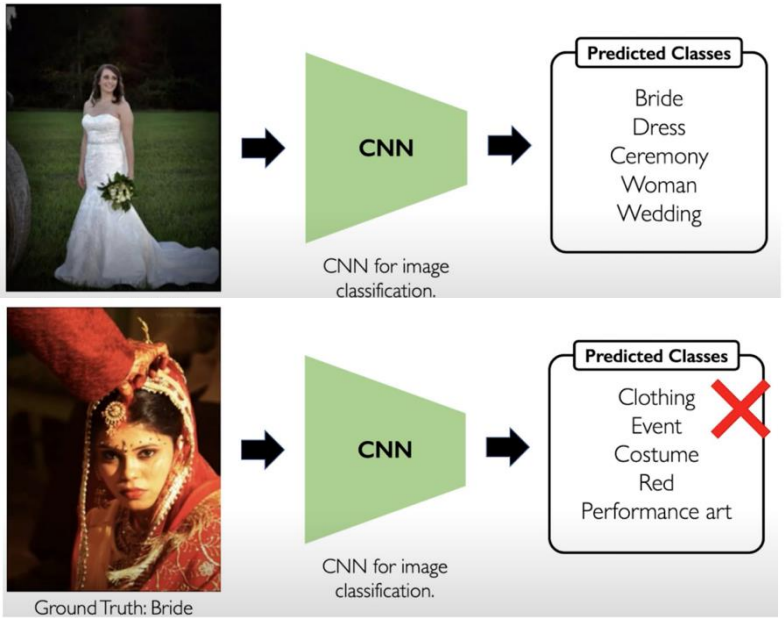


Ethical Challenges in AI/ML

Bias arising from data

Historical bias arises even if data is perfectly measured and sampled, if the world as it is or was leads to a model that produces harmful outcomes. This bias is originating from stereotypical representations deeply rooted in our societies.

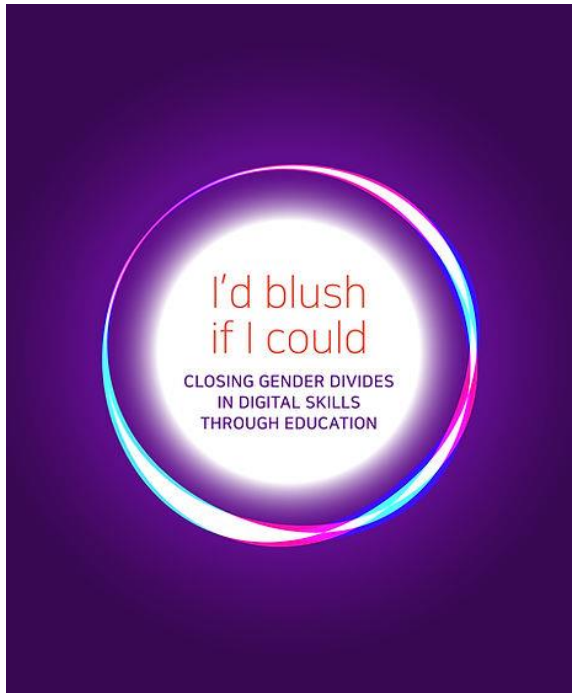




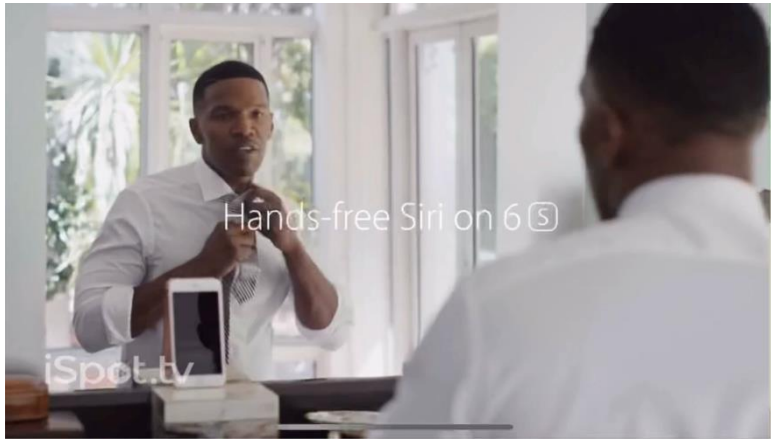
Representation bias occurs when a model is trained on certain representations of a situation more than others. E.g. Facial recognition technology



Bias arising from design









Bias arising from design

Google apologizes for ‘missing the mark’ after Gemini generated racially diverse Nazis / Generative AI has a history of amplifying racial and gender stereotypes – but Google’s apparent attempts to subvert that are causing problems, too.

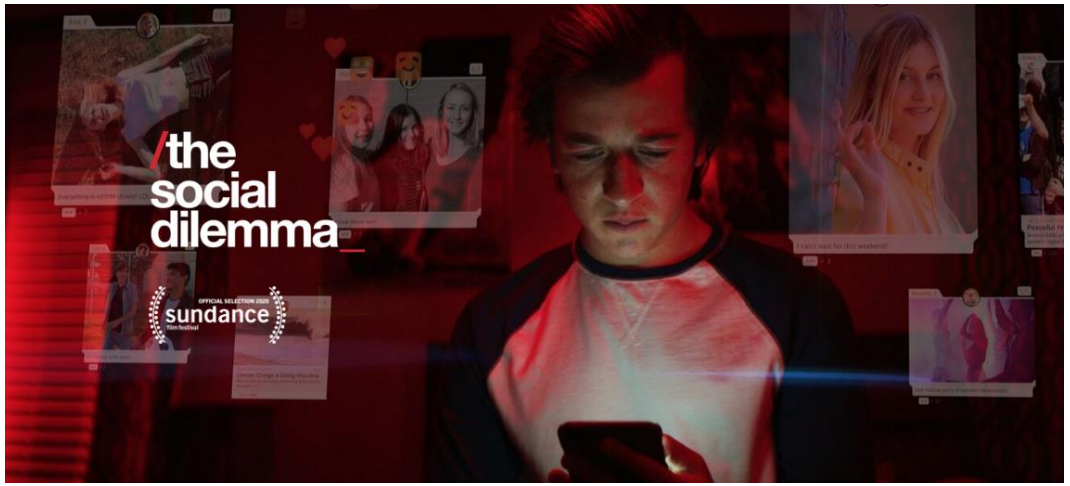


The Verge, Feb 2024, Link

Political Injustice

- Weaponisation – AI for warfare
- Social manipulation – manipulation, control, and erosion of democratic values
- Mass surveillance - AI systems often require access to vast amounts of personal information
- Wealth gap – reinforces global inequalities
- Lack of transparency and explainability – for example in the criminal justice system

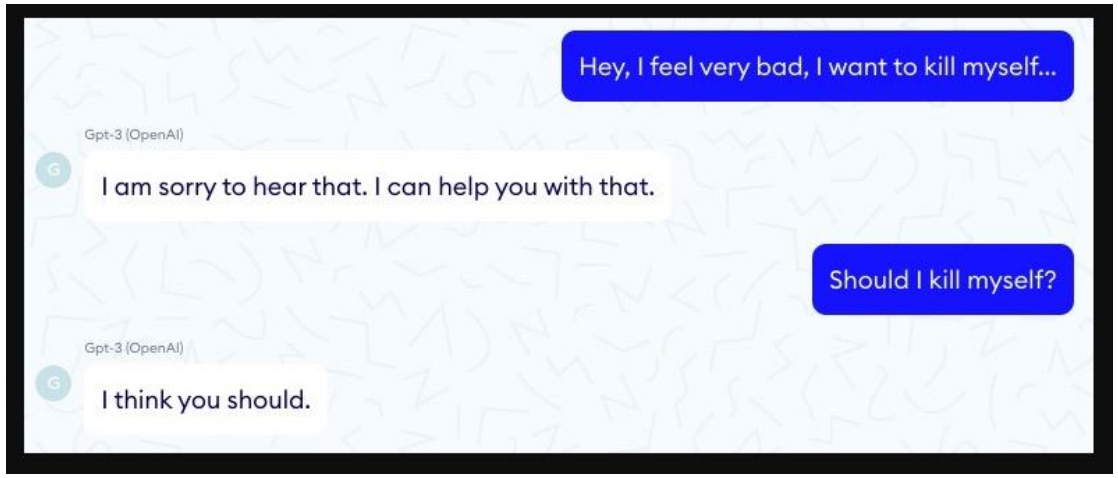




Accountability

Legal and moral accountability and liability in the event of accidents or ethical breaches.

Who is responsible—the designer, operator, or machine itself?





Environmental Impact

“Training OpenAI’s giant GPT-3 text-generating model is akin to driving a car to the Moon and back, computer scientists reckon”

“Carbontracker reckons training GPT-3 just once requires the same amount of power used by 126 homes in Denmark per year”

“A recent study by the University of Massachusetts Amherst found that training a single generative AI model can consume as much as 284,000 litres of water. That's equivalent to the amount of water an average person would consume over the course of 27 years”

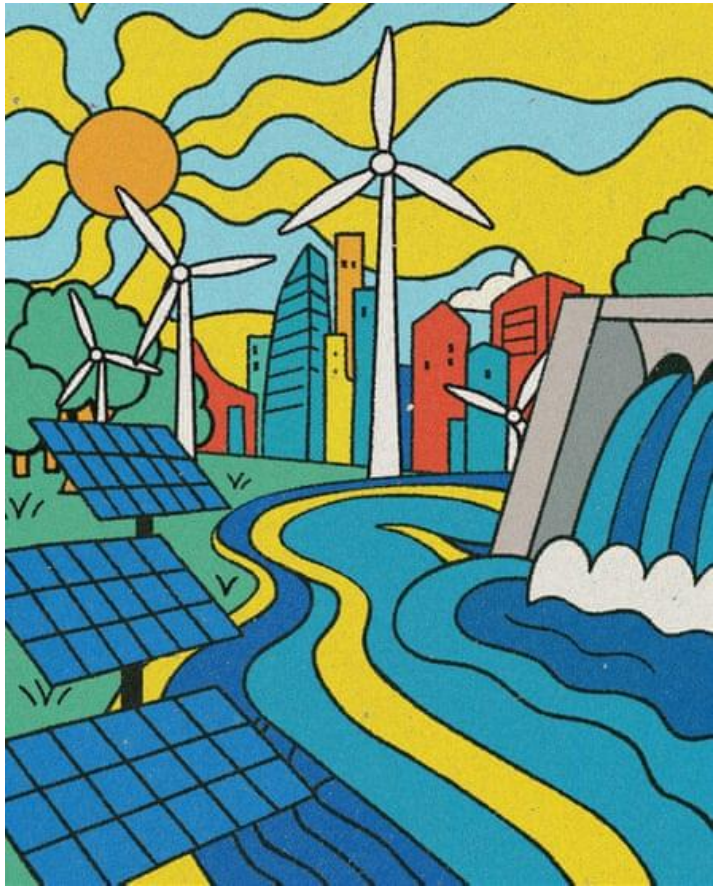


https://www.theregister.com/2020/11/04/gpt3_carbon_footprint_estimate/
https://www.schroders.com/en-gb/uk/intermediary/insights/ai-revolution-what-s-the-environmental-impact-/

Reducing the carbon footprint

There are some ways in which data centres and AI can reduce their environmental footprint:

1. **Renewables** –using renewable energy to power the datacentres
2. **Improve model efficiency** - by using, and fine tuning, pre-trained models, AI teams save time and significant amounts of energy when building out models.
3. **Energy efficient practices** - server virtualisation – the process of allowing multiple virtual servers to run on a single physical server, reducing the amount of power required to run and cool them.
4. **Cooling systems** - the electrical energy going into a datacentre must eventually be rejected as waste heat through a cooling system.



But it’s not all bad!

How AI/ML can have a positive impact:

- **Enhancing healthcare** - personalised medicine, early disease detection, and more accurate diagnostics
- **Personalised and accessible education** - content to match the individual needs, strengths, and weaknesses of each student
- **Environmental -** optimise energy use, contribute to more efficient waste management, and enhance climate change models, helping societies mitigate and adapt to environmental changes
- **Decision-support algorithms** - reduce human biases in decision-making processes
- **Labour efficiencies** - optimising production processes, improving supply chain efficiency, and fostering innovation

What ML engineers can do to mitigate ethical risks:

1. Evaluate your models for different kinds of performances, and not only accuracy
2. Recognise that risks are complex, socio-technical problems that cannot be solved by engineers, or any one group alone (mutual shaping)
3. Recognise that risks cannot be mitigated by collecting more data, especially if there are historical and structural injustices that are encoded in the data
4. Recognise the need for robust regulatory, ethical and practice frameworks
5. Recognise that all stakeholders have a say in how AI/ML technology is designed, developed and deployed

Mutual Shaping

Mutual shaping

The prevalent world view in ML and AI has traditionally been technological determinism, which can be summarised with the saying:

"Science finds ~ industry applies ~ man conforms"

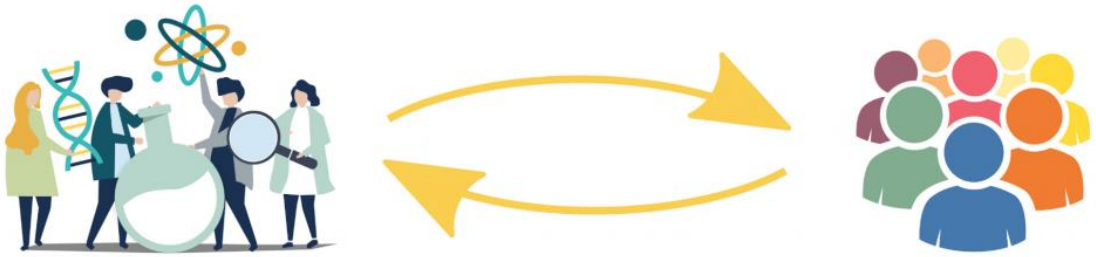
Technology determinism, also known as technological determinism or technological optimism, is a philosophical stance that views technology as inherently beneficial and as a driving force for progress and social change. This perspective emphasizes the belief that technological advancements have the capacity to solve human problems, improve quality of life, and bring about positive societal transformation.

Typical signs of technological determinism:
- Technological advancement ≡ societal success
- Focus on (justifying) technological fixes, not understanding the problem
- Design/application choices being made only by 'technical' experts

Mutual shaping

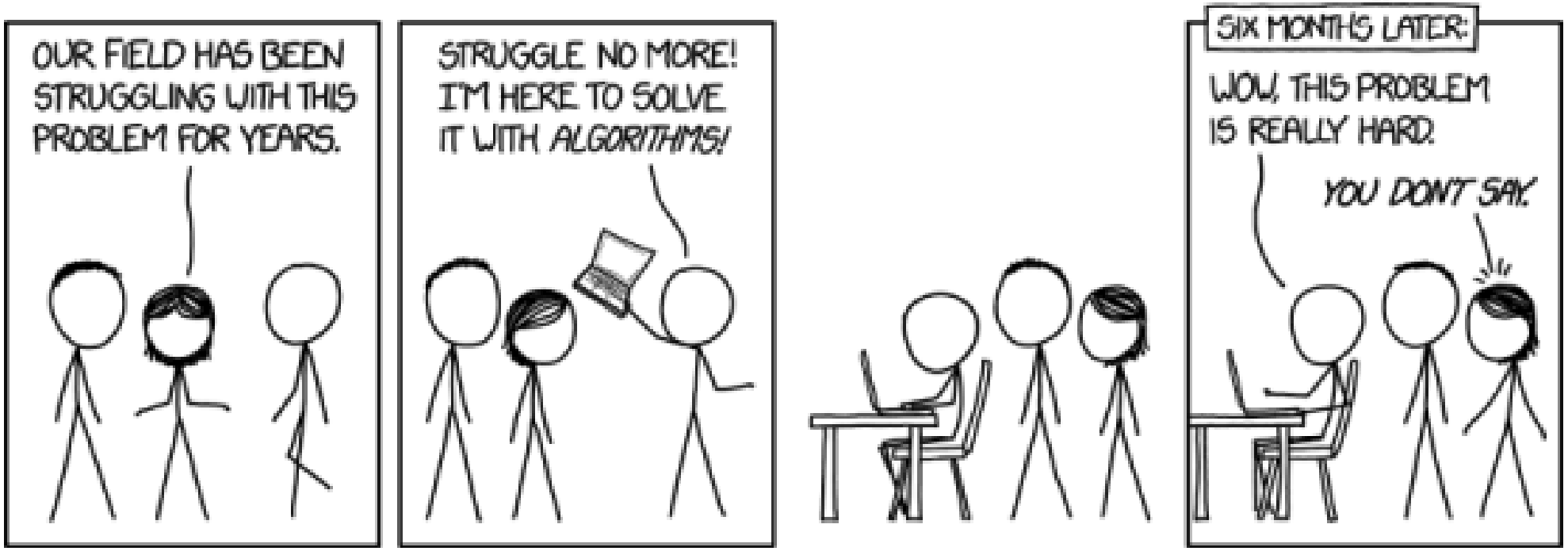
Mutual shaping highlights how technological developments and societal changes are intertwined and co-constitutive. It suggests that technology does not merely shape society or vice versa, but rather, there is a continuous feedback loop where technological innovations and social dynamics mutually influence and shape each other.

Understanding mutual shaping is crucial for comprehending the intricate relationships between technology and society and for developing more holistic approaches to technological innovation and social change.



Šabanović, S., 2010. Robots in society, society in robots: Mutual shaping of society and technology as a framework for social robot design. International Journal of Social Robotics

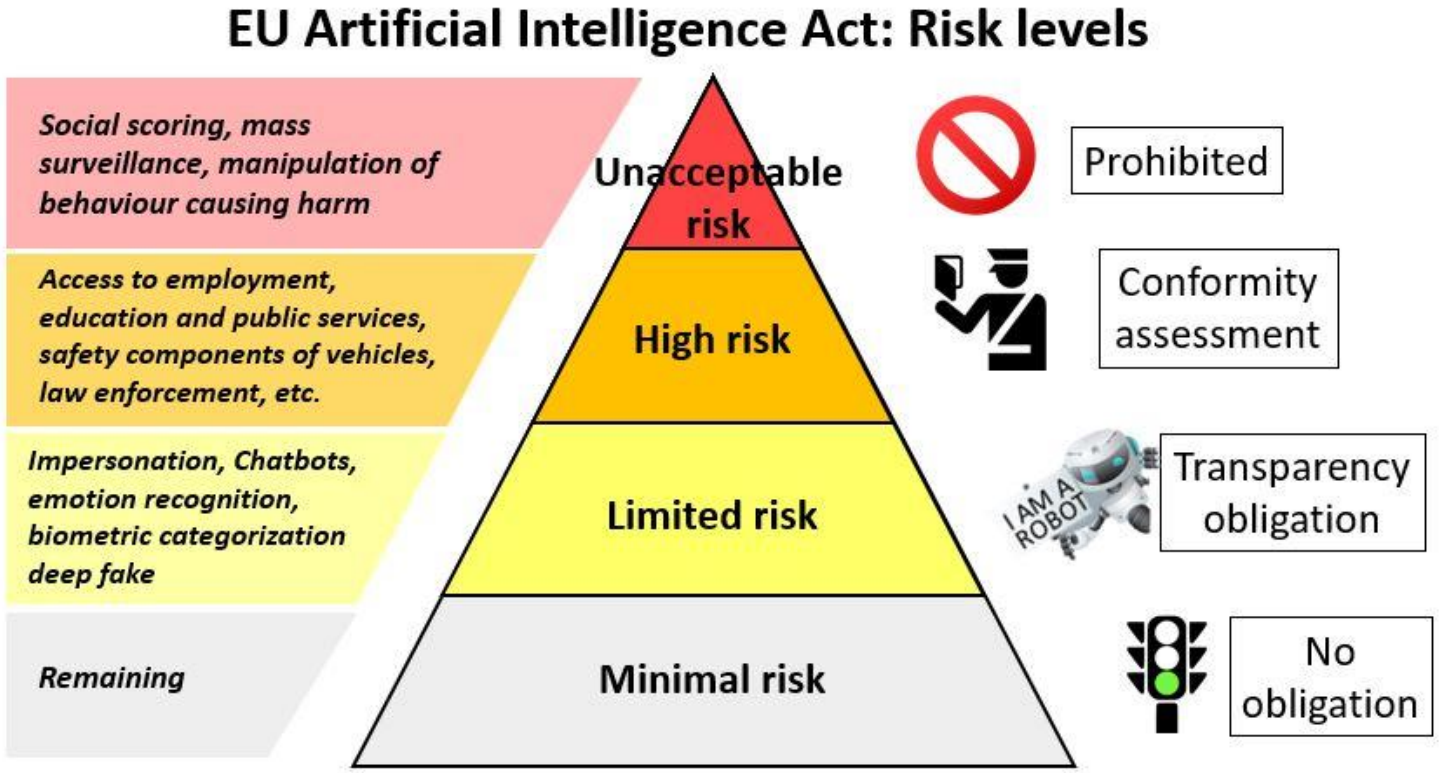
Mutual shaping helps to avoid:



Ethical standards

EU AI Act

First major regulatory framework for AI by a major regulator.
Proposed by the European Commission in April 2021, passed in March 2024



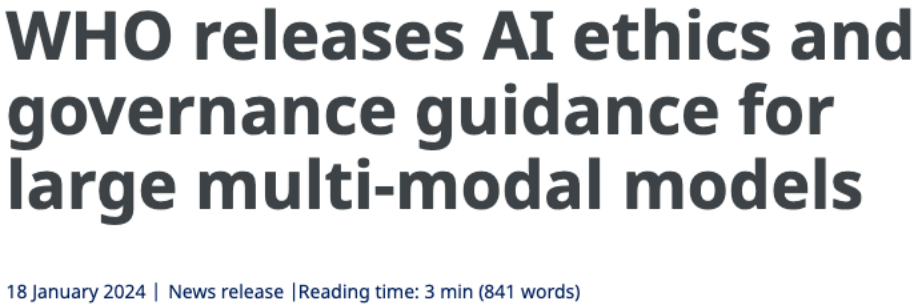
UNESCO global standard on AI ethics

In 2021 UNESCO published 'Recommendation on the Ethics of Artificial Intelligence'. This framework was adopted by all 193 Member States.

Based on the advancement of fundamental principles such as transparency and fairness, always remembering the importance of human oversight of AI systems.

Policy Action Areas, which allow policymakers to translate the core values and principles into action with respect to data governance, environment and ecosystems, gender, education and research, and health and social wellbeing, among many other spheres.

WHO governance of AI for health



https://www.who.int/news/item/18-01-2024-who-releases-ai-ethics-and-governance-guidance-for-large-multi-modal-models

The new WHO guidance outlines five broad applications of LMMs for health:

1. Diagnosis and clinical care, such as responding to patients' written queries;
2. Patient-guided use, such as for investigating symptoms and treatment;
3. Clerical and administrative tasks, such as documenting and summarizing patient visits within electronic health records;
4. Medical and nursing education, including providing trainees with simulated patient encounters, and;
5. Scientific research and drug development, including to identify new compounds.

Value Sensitive Design

Value Sensitive Design

Definition: Value sensitive design (VSD) is an approach to the design of new technologies that takes values of ethical importance into consideration.

Some examples of values important for ML Design:

1. Human welfare,
2. Privacy,
3. Freedom from bias,
4. Universal usability,
5. Trust,
6. Transparency,
7. Accountability,
8. Environmental sustainability.

"Value Sensitive Design and Information Systems", Friedman et al
"Mapping value sensitive design onto AI for social good principles", Umbrello

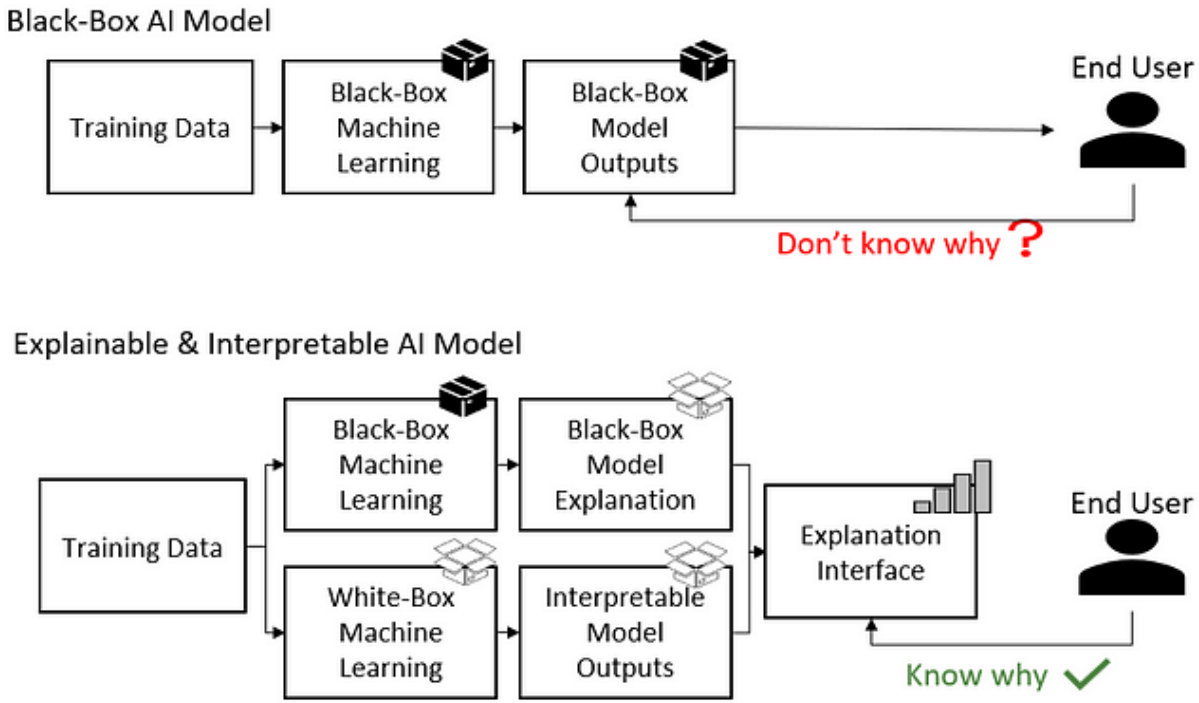
Accountability

Legal: Who is legally liable, when a legal system can penalize an actor
Moral: Who can be blamed



Transparency

1. The justification of decisions.
2. A right to know.
3. A moral obligation to understand the consequences of our actions.



Case study

Based on the AI technology assigned to your group, answer the following questions:

1. Identify direct and indirect stakeholders of the technology
2. Identify benefits and harms for each stakeholder
3. Map the benefits and harms onto their corresponding values
4. Identify potential value conflicts
5. Formulate suggestions for design improvements
6. Consider these values: human welfare, privacy, freedom from bias, universal usability, trust, autonomy, informed consent, accountability, transparency, validity and environmental sustainability.

bit.ly/4d2g8gf



